Hossein Zareinejad

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Summary

BSc student in Computer Engineering with hands-on experience in NLP, LLM evaluation, and building production-ready pipelines. Proficient in developing Retrieval-Augmented Generation (RAG) systems, implementing conversational agents, and fine-tuning foundation models for domain-specific applications. Experienced in multi-domain data curation and diagnosing model behaviors like hallucinations. Highly motivated to solve engineering challenges in scaling large language models, processing high-quality datasets, and deploying intelligent systems using clean-code practices.

Education

Amirkabir University of Technology Sep 2022 – Mar 2027 (Expected)
B.Sc. in Computer Engineering *Tehran, Iran*
GPA (last two years): **3.69 / 4.0** | Overall GPA: **3.17 / 4.0**

Talayedaran High School (Gifted High School) Sep 2019 – Jun 2022
High School Diploma *Iran*

Research Interests

Interests: Large Language Models (LLMs), RAG Architectures, Agentic Systems, LLM Benchmarking & Evaluation, Fine-Tuning, Autonomous Planning.

Professional & Research Experience

Researcher — Cross-lingual & Contextual Word Sense Induction Sep 2026 – Present
Research Project (under Dr. Masood Ghayoomi) *Tehran, Iran*

- Ported a legacy Python 2 Word Sense Induction pipeline to Python 3 and verified it against the official SemEval-2010 Task 14 scorers; evaluated MUSE/VecMap cross-lingual embedding alignment on Persian and English and found no improvement.
- Benchmarked BERT-base, ParsBERT, mBERT and XLM-R (224 configurations) against a Word2Vec baseline, raising the harmonic mean of V-measure and F-score from 36.5 to 55.9 on Persian and from 24.7 to 47.9 on English; language-specific models ranked first among base-size models.

Undergraduate Researcher — LLM Evaluation & Benchmarking Mar 2026 – Present
Amirkabir University of Technology (under Dr. Saeedeh Momtazi) *Tehran, Iran*

- Built a 5,793-question Persian multiple-choice benchmark from official Iranian exams (law, medicine, psychology, finance, Islamic studies, art), including an OCR-based PDF-to-JSON pipeline and checks of every question against the source booklets.
- Evaluated 14 open-weight LLMs (Gemma, Qwen, Llama, and Persian-adapted models) with lm-evaluation-harness and vLLM, using two zero-shot protocols: option-label scoring averaged over cyclic option permutations to remove position bias, and reason-then-score.
- Ran paired statistical analyses (McNemar with Holm correction, bootstrap CIs, multi-seed runs) and found that Persian adaptation did not improve accuracy over the unadapted base models; first-author paper in preparation.

Undergraduate Researcher — Autonomous Driving & LLM Agents Mar 2026 – Present
Amirkabir University of Technology (under Dr. Mahdi Javanmardi) *Tehran, Iran*

- Developing agentic decision-making frameworks based on generative LLMs for autonomous vehicles as an alternative to traditional rule-based behavioral planners.
- Designing integration pipelines connecting high-level LLM agent outputs with ROS2 / Autoware lower-level planning and control modules.
- Drafting simulated driving scenarios and evaluation suites for testing planning logic, informed by recent work on foundation models for robotics.

Undergraduate Researcher — NLP & Large Language Models Mar 2025 – Feb 2026
Amirkabir University of Technology (under Dr. Ehsan Nazerfard) *Tehran, Iran*

- **Event-Escalation Forecasting from Persian News:** Built an end-to-end tool (Django, React) that turns a bundle of dated Persian news items into a fixed-format 72-hour escalation assessment (risk level, key signals, per-actor behavior, ranked scenarios), running fully offline on a local Qwen2.5-7B via llama.cpp (∼45 s per bundle).
- **Reliable LLM Signal Reading:** Replaced free-form generation with five few-shot classifiers scored from token logits (averaged over 4 example orderings) plus a rule layer; found and removed KV-cache prefix reuse as the cause of up to 6-point output drift, making repeated runs byte-identical.
- **Evaluation:** Hand-labelled 402 news items and measured on held-out batches only: fully-correct labelling of unseen items rose from 47% to 74% (event type 97%, direction 84%); a 33-case historical backtest ranked the correct scenario first in 25 cases; backed by 127 automated tests.

Teaching Assistantship

Amirkabir University of Technology

- **Software Engineering II** (Head TA) — Dr. Faezeh Gohari *TA: Fall 2025, Spring 2026 | Head TA: Fall 2026*
- **Software Testing** (Head TA) — Dr. Faezeh Gohari *TA: Fall 2025, Spring 2026 | Head TA: Fall 2026*
- **Industrial Internship** (Head TA) — Dr. AmirHossein Roshanzamir *Spring 2026, Fall 2026*
- **Data Mining** — Dr. Behnam Yousefimehr *Fall 2026*
- **Mobile Application Development** — Dr. Masoumeh Taromirad *Fall 2026*
- **Advanced Programming** — Dr. Masoumeh Taromirad *Fall 2025, Spring 2026*
- **Entrepreneurship** — Dr. Morteza Zakeri *Fall 2025*
- **Logic Circuits** — Dr. Mehdi Sedighi & Dr. Morteza Saheb Zamani *Fall 2024*

Academic Projects

IdentityGraph (Multilingual Entity Resolution) | *Python, Qdrant, hazm, Docker* GitHub

- Built a record-linkage pipeline that merges records of the same entity across four heterogeneous Persian/English data sources and links each cluster to a reference record set, with per-feature evidence and calibrated confidence (logistic + isotonic regression); evaluated on synthetic data.
- Designed 8-path candidate blocking (101M → 787K pairs at 97.0% recall), Fellegi–Sunter pairwise scoring, and one-to-one constrained clustering (Hungarian algorithm), raising B-cubed F1 from 0.675 to 0.889.
- Reached 82.2% top-1 / 92.5% top-5 accuracy on 1,752 unseen clusters (91.4% precision at 85.2% coverage); fixing transliteration and phonetic keys raised cross-script name retrieval on 7,812 real Wikidata name pairs from 66.1% to 90.0%.

GraphNLP (Text Summarizer & Entity Graph) | *Python, FastAPI, SpaCy, Hugging Face* GitHub

- Built a FastAPI web app that summarizes long articles abstractively (Hugging Face BART) and extractively (SpaCy).
- Implemented Named Entity Recognition and relation extraction, rendering entities and their relationships as an interactive, physics-simulated Vis.js graph.
- Added click-to-focus entity highlighting and a Hugging Face dataset loader streaming articles from cnn_dailymail.

FleetSim MAPF (Multi-Agent Path Finding) | *Python, Pygame* GitHub

- Built a Multi-Agent Path Finding simulator using Conflict-Based Search (CBS) with a 3D Space-Time A* low-level planner for optimal, collision-free routing across agents.
- Added interactive dynamic re-planning: agents recompute collision-free paths in real time as obstacles are placed or removed mid-simulation.
- Built a Pygame GUI with sub-pixel LERP animation and live CBS diagnostics, supporting official Moving AI benchmark maps and scenarios.

MelodyChat (Sickimfy) | *Kotlin, Jetpack Compose, Ktor, WebSockets, Room* GitHub

- Developed a full-stack Android music app featuring streaming, offline downloads, and real-time chat.
- Built the UI using Jetpack Compose (MVVM) and integrated AndroidX Media3 for background playback.
- Engineered a Ktor backend with WebSocket chat hubs, JWT authentication, and Room local persistence.

Technical Skills

Core Programming: Python, C, Kotlin, JavaScript (ES6+), SQL, Linux CLI, Git.
AI & LLM Ecosystem: PyTorch, Hugging Face Transformers, vLLM, llama.cpp, lm-evaluation-harness, Sentence Transformers, spaCy, Fine-Tuning, RAG Pipelines, Vector Databases (Qdrant).
Models: LLaMA, Qwen, Gemma, Mistral, BERT, XLM-R, ParsBERT.
Evaluation & Data: LLM Benchmarking, Statistical Significance Testing (McNemar, Holm Correction, Bootstrap CIs), OCR-based Data Pipelines, Text Preprocessing.
Tools & Frameworks: Django, FastAPI, REST APIs, GraphQL, React.js, Next.js, PostgreSQL, Docker, ROS2, Autoware.

Training Courses

- **Machine Learning & Deep Learning:** Supervised Machine Learning (Coursera/Andrew Ng), Advanced Learning Algorithms (Coursera/Andrew Ng), Machine Learning (Sharif University of Technology / Ali Sharifi-Zarchi).
- **Web Development:** Task-Oriented Front-end Development Bootcamp (Coursera).

Languages

Persian: Native **English**: Full Professional Proficiency